

Draft genome annotation of the heat tolerant fungi *Talaromyces trachyspermus* isolated from fruit juice

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Abstract: The annotation of fungal genomes is essential for understanding biological traits such as spoilage mechanisms in food systems. This project aims to analyze the genome of a fungus isolated from Portuguese fruit juice matrixes which presents the ability to tolerate heat and also produce mycotoxins, by using bioinformatics tools. The genome sequence of *Talaromyces trachyspermus* was obtained using paired-end, short-read sequencing obtained from Illumina technology. The reads (FASTQ files) were submitted to quality check by using FASTQC, followed by pre-processing using Trimmomatic to remove low quality sequences. Pre-processed files underwent a final quality check and were then assembled by using Spades. The quality assessment of the assembly was performed with Quast. Finally, the contigs were annotated with AUGUSTUS model database. A genome of 31Mbp with 441 contigs, 47.77 %GC content, and N50 of 343 kbp was obtained. Annotation against a database containing the proteins of all *Talaromyces* species whose genomes are available, resulted in the identification of 11441 unique proteins that can be expressed by the *Talaromyces* isolate. Of those, at least 35 are related to mycotoxins production, namely aflatoxin, and 65 are linked to heat tolerance (hsp90, hsp70 and calcineurin) with the vast majority most similar to the proteins found in *Talaromyces atrovirens*. These results confirm the ability of this isolate to produce mycotoxins and resist in environments at high temperatures, since it has the genomic machinery associated with those phenotypes.

Keywords: *Talaromyces trachyspermus*, pipeline, fungi, fruit juice, genome annotation, assembly

1. Introduction

The preservation of pasteurized fruit products is mainly based on a combination of their high acidity, the heat and pressure process applied (Rico-Munoz et al., 2017). Although this is usually sufficient to eliminate and prevent the growth of most spoilage microorganisms and pathogens, it is not sufficient to eliminate the spores of some microorganisms (Dos Santos et al., 2018; Tranquillini et al., 2017). One group of microorganisms that poses a particular challenge are heat-resistant molds (HRMs), which are able to survive the pasteurization process (Rico-Munoz et al., 2017). Some of the main HRMs found in fruit juices are *Neosartorya fumigata*, *Neosartorya fischeri*, and *Byssoschlamys nivea*, and *Talaromyces trachyspermus* (Dos Santos et al., 2018; Rico-Munoz et al., 2015; Houbaken et al., 2008). It is important to note that there is a plethora of other HRMs that are not noted, with many being found in the wild in association with fruits (Rico-Munoz et al., 2017).

Talaromyces trachyspermus was first identified as a heat-resistant mold (HRM) in 1993, when it was observed growing within a mixed culture isolated from fruit juice after being exposed to 80°C heat for 30 minutes in an autoclave (Enigl et al., 1993). Its classification as an HRM was later reaffirmed by Tranquillini et al. (2017), who investigated the species' D-values.

One of the main types of heat-resistant spores are ascospores, which can be induced by stressful situations (Houbaken et al., 2008). Ascospores are specialized thick-walled spores produced during the sexual cycle of ascomycete fungi, providing them with remarkable resistance to heat and harsh environmental conditions (Houbaken et al., 2008; Rico-Munoz et al., 2017). This sexual cycle involves the fusion of compatible mating types, leading to meiosis and the formation of ascospores within asci (Rico-Munoz et al., 2017). Unlike the asexual cycle, which rapidly propagates fungi through conidia, the sexual cycle enhances genetic diversity and enables the development of stress-resistant structures (Houbaken et al., 2008). The interplay between the asexual and sexual cycles ensures both rapid

colonization and long-term survival, making these fungi persistent contaminants in food products, particularly fruit juices (Houbraken et al., 2008; Rico-Munoz et al., 2017).

The *Talaromyces* genus has 97 entries in NCBI (as of 20/3/25) and only a single genome assembly of *T. trachyspermus* (CM035349.1). The isolate was an endophyte of *Withania somnifera* (Sahu et al., 2019). The background of our isolate, coming from a fruit juice matrix of Portuguese origin, is vastly different and most likely of separate strains than the isolate mentioned above. When checking UniProt, although the genus has around 105,000 entries, only 32 entries are of *T. trachyspermus* proteins (Taxonomy ID: 28566) (as of 20/3/25).

Consequently, annotating the genome of *T. trachyspermus* is highly significant, as gaining a deeper understanding of this fungus could lead to effective strategies for eradicating it from food products. This would not only help fruit juice manufacturers minimize spoilage-related waste but also improve consumer safety at the point of purchase. Additionally, other industries that consider this genus a promising source for metabolite production (Lan & Wu, 2020; Zhai et al., 2016) could also reap the benefits of this research.

1.1 Whole-Genome Sequencing (WGS)

Whole-genome sequencing (WGS) has revolutionized fungal research by providing comprehensive genetic insights into thousands of fungal species (Ma & Fedorova, 2010; Fitzpatrick, 2012). As a cost-effective and high-throughput method, WGS enables researchers to decode entire fungal genomes, revealing their evolutionary relationships, pathogenic mechanisms, and metabolic capabilities. The advent of next-generation sequencing (NGS) platforms like Illumina, PacBio, and Oxford Nanopore has made fungal genome sequencing more accessible, leading to fungi becoming some of the most well-sequenced eukaryotes (Haridas et al., 2011; Ma & Fedorova, 2010; Fitzpatrick, 2012). This extensive genomic data has facilitated advancements in fungal taxonomy, biotechnology, and disease management, enhancing our understanding of their roles in ecosystems and industries (Fitzpatrick, 2012).

The Illumina technology was chosen to sequence the genome of the *T. trachyspermus* strain isolated from the fruit juice matrix. The Illumina sequencing platform relies on highly accurate short-read technology, starting with DNA extraction and library preparation. The extraction process involves isolating high-quality genomic DNA from a sample, typically using commercial kits or phenol-chloroform methods (Shendure et al., 2008). The extracted DNA is then fragmented into short sequences, usually between 150–300 base pairs, through enzymatic or mechanical shearing. Adapters are ligated to these fragments, allowing them to bind to the sequencing flow cell (Quail et al., 2012). During sequencing, the fragments undergo bridge amplification, creating clusters of identical DNA strands (Bentley et al, 2008). With it being the method used for the isolation of the *T. trachyspermus* strain, it is important to adapt our pipeline to fit such needs.

1.2 Overview of Genome Annotation

Genome annotation involves multiple computational steps to extract biologically meaningful information from raw sequencing data. These steps include quality control, genome or transcriptome assembly, structural gene prediction, and functional annotation. Each phase requires different types of input data, applies distinct computational principles, and outputs intermediate files that feed into subsequent stages (Yandell M et al 2012).

Preprocessing

Preprocessing focuses on the quality assessment and cleaning of DNaseq reads, which are typically obtained in FASTQ format from next-generation sequencing platforms like Illumina (Mardis, E.R 2008). Each read contains a nucleotide sequence and an associated Phred quality score that reflects the confidence of each base call. However, raw reads may contain sequencing adapters, low-quality bases, or contaminants.

To address these issues, FastQC (Andrews et al., 2010) can generate graphical summaries of sequencing quality. Including per-base quality, guanine and cytosine content (GC content), sequence duplication levels, and overrepresented sequences, which helps to identify problematic regions. Subsequently, Trimmomatic (Bolger et al., 2014) can remove adapter sequences and trim or discard low-quality reads. Trimmomatic is especially valued for its flexibility, supporting single-end and paired-end reads and allowing parameter customization to suit different datasets. The result is a set of cleaned, high-quality reads, in FASTA files, suitable for downstream analyses (Bolger et al., 2014).

Genome Assembly

Next comes genome assembly, where the previously cleaned reads are ordered to construct a possible genome stored in a BAM file (Li et al., 2009). A BAM file (Binary Alignment/Map) is a binary, compressed version of a SAM file (Sequence Alignment/Map), which stores alignment information for sequencing reads mapped to a reference genome. Many tools require the BAM format for downstream predictions due to its computational efficiency.

Repetitive elements are particularly abundant in eukaryotic genomes (Yandell & Ence, 2012). To mitigate their impact on sequence analysis, masking is performed by replacing repetitive sequences with placeholder characters, thereby preventing false alignments during downstream processing. Tools such as RepeatModeler2 (Flynn et al., 2020) enable *de novo* identification of transposable elements and other repeats, aggregating them into a custom repeat library. This library can then be used as input for RepeatMasker (Chen N. 2004) which systematically masks these regions. Alternatively, RepeatMasker can utilize pre-defined repeat libraries available for a wide range of organisms. This masking step is crucial for reducing errors in gene prediction and other genomic analyses, ensuring more accurate annotation and interpretation of eukaryotic genomes (Yandell & Ence, 2012).

There are two main strategies for genome assembly. *De novo* assembly, which predicts genome structure using computational algorithms from scratch, or reference-based assembly, leverages pre-existing genomic data from related species (Gorman et al., 2023).

In cases where reference genomes are incomplete or absent, as is common for non-model fungi, guided transcriptome assembly is particularly valuable (Ungaro et al. 2017). In this method, DNA-seq reads are first aligned to a draft genome using spliced aligners such as HISAT2, which are optimized to recognize exon-intron boundaries in eukaryotic transcripts (Kim et al., 2015). These alignments guide the assembly of full-length transcripts using Trinity (Grabherr et al., 2011), a robust *de novo* assembler that reconstructs isoforms and transcripts even in species lacking well-annotated genomes. Trinity uses a three-step process involving Inchworm, Chrysalis, and Butterfly modules, which sequentially assemble reads into contigs, cluster them into graphs, and reconstruct alternative splice forms and transcript variants (Haas et al., 2013).

Gene Prediction or Annotation

Gene prediction involves identification of gene structures in the previously assembled genome. For eukaryotes, this includes exons, introns, and UTRs. GFF files (also known as GFT files), or Gene Feature Format files, store the structural information gathered from gene prediction for later functional annotation (Zhang, X et al 2020).

Gene annotation can be achieved using two main approaches. *Ab initio* methods rely on statistical models trained on known gene features, such as AUGUSTUS (Stanke et al 2006) GeneMark-ET (Lomsadze et al., 2014), and SNAP (Korf, 2004). For example, AUGUSTUS uses a generalized hidden Markov model for exon-intron predictions and can function independently or as part of larger pipelines (Stanke et al., 2006). It is particularly effective when complemented by transcriptome evidence, enabling the utilization of limited information for organisms like *T. trachyspermus*. Evidence-based approaches leverage DNA-seq alignments and homology to known proteins to enhance predictions.

Pipelines like MAKER (Cantarel et al., 2008) and BRAKER2 (Brůna et al., 2021) combine these methods to produce accurate gene models. MAKER integrates repeat masking, aligns transcriptomic and protein data, and incorporates tools like SNAP and AUGUSTUS. Similarly, BRAKER2 incorporates DNA-seq data with *ab initio* tools, like AUGUSTUS and GeneMark-ET, to refine predictions.

Functional Annotation

Once genes are predicted, their biological functions can be inferred using a combination of computational tools (Yandell, M et al 2012). Functional annotation involves multiple steps to assign roles to genes, assess genome quality, and gain insights into metabolism, resistance, and pathogenesis. Many different strategies and tools can be utilized to obtain functional information from genomic data.

For instance, predicted protein sequences can be aligned against the NCBI non-redundant (nr) protein database using BLASTp (Altschul et al., 1997). This comparative approach leverages evolutionary conservation by identifying homologous sequences in well-characterized organisms. High-scoring matches with low E-values provide strong candidates for functional assignment, enabling the identification of gene families, protein functions, and enzymatic activities. This approach is particularly valuable for characterizing metabolic pathways and resistance mechanisms (Krijgsheld, P et al 2013).

BUSCO (Benchmarking Universal Single-Copy Orthologs) is an example of a tool which evaluates the completeness of genome assembly and annotation by searching for conserved single-copy orthologous genes (Simão et al., 2015). These orthologs are essential functional components expected to be present in nearly all fungal genomes. Metrics such as the presence, duplication, fragmentation, or absence of these genes provide quantitative insights into genome quality. For *T. trachyspermus*, the fungal lineage dataset (e.g., `ascomycota_odb12`) ensures phylogenetic relevance in assessing annotation accuracy.

InterProScan is a tool which integrates multiple signature recognition methods (e.g., Pfam, SMART, TIGRFAMs, SUPERFAMILY) to identify conserved protein domains and classify proteins into families (Jones et al., 2014). By assigning Gene Ontology (GO) terms and detecting structural motifs, it informs us of biochemical functions, cellular roles, and evolutionary relationships. In fungi, this approach provides insights into enzymes associated with secondary metabolism, cell wall synthesis, environmental adaptation, and others. These findings can further inform efforts to understand the underlying mechanisms of fungicide and heat resistance in *T. trachyspermus*, thereby enabling the development of more effective control strategies. Additionally, they can aid in designing targeted interventions to inhibit the presence and proliferation of *T. trachyspermus* in unwanted environments, such as agricultural settings, storage facilities, and processing areas. Other bioinformatics tools exist to identify protein domains, such as, reCOGnizer, which uses as reference more than 7 databases (Sequeira et al 2022).

1.3 Annotating a Poorly Studied Species

When annotating a species with very little prior information, standard annotation pipelines relying on well-characterized reference genomes may not be suitable. Instead, a combination of de novo gene prediction and DNA-seq evidence is crucial for improving accuracy (Granados-Casas et al., 2023). Since existing gene models may not be applicable, multiple tools should be used to ensure comprehensive prediction, and functional annotation must leverage broader taxonomic databases rather than species-specific ones (Quan et al., 2024).

Objective

The primary goal of this study is to analyze the genome sequence of a *Talaromyces trachyspermus* strain isolated from Portuguese fruit juice matrices using bioinformatics tools. The focus is to identify genomic features associated with key phenotypes, such as mycotoxin production and heat tolerance. To achieve this, a review of existing bioinformatics

resources was conducted to develop a tailored workflow for analysis.

2. Methodology

Pipeline for Functional Annotation

In this study, we implemented a genome annotation pipeline specifically tailored for fungal genomes (Fig. 1). This pipeline integrates all major steps necessary for accurate annotation, including de novo assembly, ab initio and evidence-based gene prediction, and functional annotation. Widely adopted tools were selected and adapted to the unique characteristics of eukaryotic, and particularly fungal genomes. Each step was applied to annotate the genome of *Talaromyces trachyspermus*, with preprocessing and quality control steps introduced to enhance the reliability of the results.

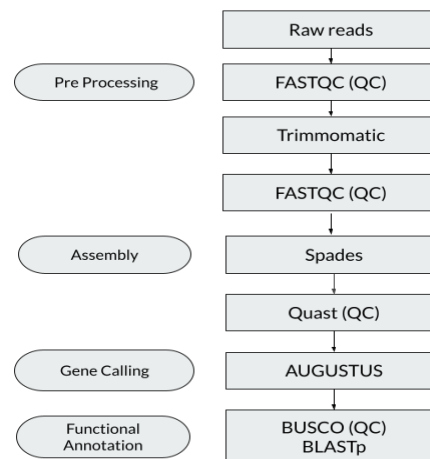


Fig. 1 – Pipeline used in this study. Quality control steps are marked with (QC)

DNA-seq preprocessing began with Illumina short reads, for which quality assessment was performed using FastQC. Low-quality reads and adapter sequences were removed using Trimmomatic with parameters optimized for Illumina-generated DNA-seq data. The cleaned reads were used for de novo assembly with SPAdes under default parameters. Assembly quality was evaluated using QUAST, applying the `--fungus`, `--eukaryote`, and `--gene-finding` options.

Gene prediction was performed using AUGUSTUS (Stanke et al., 2006), with the resulting scaffolds as input. As AUGUSTUS requires a reference species for ab initio prediction, four closely related *Aspergillus* species were considered based on their availability in the AUGUSTUS model database. Initial attempts using *Aspergillus fumigatus* did not yield usable results; however, successful predictions were obtained with *Aspergillus oryzae*, which was subsequently used for all analyses. AUGUSTUS was run with the following parameters `--uniqueGeneId=true`, `--singlestand=true`, `--gff3=on`, `--species=aspergillus_oryzae`, `--stopCodonExcludedFromCDS=false`, `--softmasking=1`, and `--trnaslation_table=12`. These settings follow those used in the FUNGAP pipeline (Min et al., 2017), which is optimized for fungal genome annotation. (?)

The resulting GFF3 files were converted to FASTA format and translated into protein sequences. To assess the quality of gene prediction, BUSCO was used to identify conserved orthologs using the `ascomycota_odb12` lineage dataset.

For functional annotation, a local BLASTp search with cut-off E-value of $1e-5$ was performed against all available *Talaromyces* protein sequences downloaded from UniProt (as of 16/06/2025). This provided initial functional hits for *T. trachyspermus* proteins.

Comparison Between Species

A second BLASTp run was conducted using the parameters `--max_target_seqs 1` and `--max_hsps 1`, to try and make each query match a single best-hit (based on the lowest E-value) in the UniProt database with all available *Talaromyces* proteins. The resulting accession numbers were mapped using UniProt's ID mapping tool, generating a comprehensive dataframe that included more selected metadata variables per match. Allowing only one entry per accession ID.

To identify proteins of interest, primarily those associated with mycotoxin production and heat resistance, a list of relevant keywords was used (mycotoxin, aflatoxin, ochratoxin, patulin, citrillin, fumonisin, and trichothecene). Keywords associated with heat resistance were hsp70, hsp90, hsp30, and calcineurin. In R, rows of the Uniprot mapped dataframe were extracted for each keyword by searching within the Protein names and Gene Ontology columns. This allowed for the generation of a curated table containing relevant protein information for *T. trachyspermus* and closely related *Talaromyces* species for each keyword.

3. Results

For the following section all, supplementary material including R scripts, extended results and tables can be found at https://github.com/samildo/Sami_Projeto25

The initial run of FASTQC indicated that the reads passed all the tests. Overall quality was very satisfying for both forward and reverse read files which had basically identical results. Even so, a Trimmomatic run was executed mainly to eliminate potential repeats and low-quality reads. With a 99% survival rate. FASTQC was run once more to check the quality of the trimmed reads, with both passing all tests.

QUAST's quality metrics of the Spades assembly yielded a genome of 31.04 Mbp and 47.77% GC content. The full assembly contains 441 contigs, an N50 of 343318 bp and an L50 of 27 with the largest being 1134506 bp.

AUGUSTUS's `gff3` output was transformed into a fasta and translated into protein sequences. Approximately 12 thousand proteins were found. Being in line with the 9-13 thousand proteins in annotated *Talaromyces* species (refer to Table 1). BUSCO analysis indicated 93.1 % completeness where out of 2826 total BUSCO groups searched, 2630 are complete of which 21 are duplicates and 2609 are single copy. Forty-five BUSCO groups are fragmented. Gene predictions were blasted without restriction on hit numbers and an E-value of $1e-5$ against the 104,170 proteins in the UniProt database using local BLASTp. Yielding 956516 hits and 2256 no hits. The number of hits per accession number was obtained by conducting a search with regular expressions on the Blastp output file via command line.

The second blast run yielded 12341 hits from the original 12439 protein sequences available as queries. With 11441 unique accessions IDs, leaving 998 queries without a hit and 900 duplicates. The mapping of the accession numbers into Uniprot's Mapping ID tool produced a data frame with 11441 entries containing a unique accession ID. The keywords search results can be found on supplementary Table S1. The keywords mycotoxin, aflatoxin and trichothecene produced respectively 25, 7 and 3 hits. The other mycotoxins did not have any results. Whilst heat, hsp70, hsp90, hsp30, and calcineurin produced respectively 35, 11, 9, 0, and 13 hits (Table 2).

Table 1. Overview of key genome assembly metrics for *Talaromyces* species available under the Reference Genome category in NCBI (taxonomy ID 5094). Metrics include genome size, contig N50, number of contigs, and GC content. As well as highlighting

two *Talaromyces trachyspermus* assemblies (strains 4014 and IRAN 3054C) alongside the assembly of *Talaromyces talaromyces* isolated from Portuguese fruit juice matrix.

Organism Name	Assembly Accession	Genome size (Mb)	Number of contigs	Contig N50	Contig L50	GC %	Assembly level	Genes
<i>T. marneffei</i>	GCA_009556855.1	34.7	1345	63.9 kb	152	46	Scaffold	10066
<i>T. rugulosus</i>	GCF_013368755.1	35.8	6	5.9 Mb	3	47	Complete Genome	11977
<i>T. stipitatus</i> ATCC 10500	GCA_000003125.1	35.7	960	897.4 kb	12	46	Scaffold	12574
<i>T. amestolkiae</i>	GCA_001896365.1	33.7	288	936.8 kb	10	46.5	Scaffold	10408
<i>T. atrovirens</i>	GCA_001907595.1	30.9	167	376.1 kb	26	44.5	Scaffold	9523
<i>T. proteolyticus</i>	GCA_021365285.1	37.6	30	2.6 Mb	6	45.5	Contig	13667
<i>T. islandicus</i>	GCA_000985935.1	34.7	1345	63.9 kb	152	46	Scaffold	10094
<i>T. trachyspermus</i>	NA	31.3	441	343.3 kb	27	47.77	Scaffold	12439
<i>T. trachyspermus</i> 4014	GCA_020137715.1	32	14	3.8 Mb	3	47	Contig	NA
<i>T. trachyspermus</i> IRAN 3054C	GCA_049901525.1	31.3	680	280.7 kb	34	47.5	Scaffold	NA

Table 2. Summary of keyword-based hit counts grouped by *Talaromyces* species. Keywords were categorized into two functional groups: Mycotoxin (including “mycotoxin,” “aflatoxin,” and “trichothecene”) and Heat-related (including “heat,” “HSP90/70,” and “calcineurin”). Hits represent the number of BLASTp accession numbers successfully mapped to UniProt entries, with keyword matches found in either the Protein Name or Gene Ontology annotation fields. Only the top BLAST hit per query was retained based on the lowest E-value. The number of hits per keyword for each species is detailed in supplementary Table S2. Organism refers to the species with the closest sequence of identity to the *T. Trachyspermus* queries.

Organism	Cumulative keyword hits	“Mycotoxin” hits	“Heat” hits
<i>Talaromyces atrovirens</i>	69	18	51
<i>Talaromyces stipitatus</i> (strain ATCC 10500 / CBS 375.48 / QM 6759 / NRRL 1006) (<i>Penicillium stipitatum</i>)	9	4	5
<i>Talaromyces proteolyticus</i>	8	6	2
<i>Talaromyces islandicus</i> (<i>Penicillium islandicum</i>)	4	3	1
<i>Talaromyces rugulosus</i> (<i>Penicillium rugulosum</i>)	4	2	2
<i>Talaromyces marneffei</i> (strain ATCC 18224 / CBS 334.59 / QM 7333) (<i>Penicillium marneffei</i>)	3	0	3
<i>Talaromyces pinophilus</i> (<i>Penicillium pinophilum</i>)	3	1	2
<i>Talaromyces amestolkiae</i>	1	0	1
<i>Talaromyces marneffei</i> PM1	1	1	0
	102	35	67

4. Discussion

The Illumina DNA-seq reads were of high quality and free of adapter contamination prior to preprocessing. As such, the application of Trimmomatic had a minimal effect on improving read quality. This aligns with the findings of Bolger et al. (2014), who showed that when input reads are already of high quality, the impact of trimming on downstream analyses is marginal. Nevertheless, the use of Trimmomatic remains important, as it plays a crucial role in improving data quality in cases where low-quality bases or adapter sequences are present Bolger et al. (2014).

The assembly falls within the expected range for previous *Talaromyces trachyspermus* assemblies (see table 1), as all available strains including IRAN 3054C (31.3 Mb) and 4014 (32 Mb) have similar genome sizes. The guanine and cytosine (GC) content is also consistent with known values for this species, with values of 47.5% and 47% reported for IRAN 3054C and 4014 strains, respectively. BUSCO analysis of the gene prediction indicates high completeness (93.1%), which is at the upper end of expectations for non-model organisms (Seppey et al., 2019). The results also show very few duplicates and missing fragments, which further supports the quality of the assembly and annotation.

Overall, the assembly can be considered satisfactory, as downstream analyses yield good BUSCO results. However, there is room for improvement in assembly contiguity. The 4014 strain has a notably more contiguous assembly, whereas the current assembly is more fragmented. Future efforts could focus on reducing the number of contigs and increasing the N50 value to improve genome quality.

Gene prediction results are within the typical range for annotated *Talaromyces* species, which generally contain anywhere between 9,000 and 13,000 genes (refer to Table 1). While previous studies suggest a direct correlation between genome size and the number of genes in fungi (Zaccaron et al., 2024), it is important to note that the reported genes count may include paralogs and should be interpreted with caution. Additional runs and independent validation are recommended to confirm the gene count and to ensure annotation accuracy.

One of the potential ways of improvement is the addition of a repeat masking step before the assembly as it is advised in fungal genome annotation pipelines (Min B et al 2017) and pipelines followed for *Talaromyces* that are annotated (Luo, J et al 2025)

The abundant number of BLASTp hits in the unrestricted run is unsurprising given that each query protein can match multiple homologous sequences, including isoforms, paralogs, and closely related proteins within the *Talaromyces* genus (Altschul et al., 1990). This reflects the natural redundancy and diversity in protein sequences both within *Talaromyces* species and in public databases like UniProt, consistent with the widespread occurrence of gene duplication and protein family expansion in fungi (Wapinski et al., 2007; Galagan et al., 2005). Consequently, the high hit count demonstrates that many *T. trachyspermus* proteins have multiple homologs in related species, indicating evolutionary conservation and potential functional redundancy, as is typical for the genus (Yilmaz et al., 2014).

The second BLAST run was designed to identify the most significant match for each query sequence by restricting the output to only the single best hit with the highest E-value. This approach helps clarify which proteins in the query are most similar to those in the UniProt database. However, duplicate accession numbers may still arise due to the nature of BLAST's reporting mechanism, even when parameters such as `-max_hsps 1` and `-max_target_seqs 1` are used. Overall, nearly all of the significant hits belonged to *Talaromyces atrovirens*, indicating that this species is the closest well-annotated relative of the query in terms of both sequence similarity and genome size. These findings suggest that *T. atrovirens* could serve as a valuable reference for future annotation and comparative genomics studies, although further testing may be required to confirm the suitability of this annotation approach for the query organism.

Aflatoxin regulatory protein domain-containing protein and Zn(2)-C6 fungal-type domain-containing protein both are involved in regulating genes for aflatoxin production (Smith C. A, et al 2007 ; Fernades, M et al 1998). Aflatoxin B1 aldehyde reductase member 3 is involved in the metabolism or detoxification of aflatoxin B1 aldehyde, a reactive

intermediate in aflatoxin biosynthesis or degradation (Johnson D.N, et al 2008). MFS aflatoxin efflux pump is involved in the efflux (active transport) of aflatoxin or related compounds out of the cell (Chanda, A et al 2010). Trichothecene 3-O-acetyltransferase-like N-terminal domain-containing protein is involved in acetylation of the C-3 hydroxyl group of trichothecene mycotoxins, thereby reducing their toxicity (Yang et al. 2023).

HSP90/70/30 were chosen as they have been previously identified to have significant differential expression under stress or heat in *Talaromyces* species (Jundra, N et al 2023, He, X et al 2021). Proteins associated with the “heat shock” keywords (Hsp70 and Hsp90) span a diverse array of molecular functions beyond classical chaperone activity (Kim et al., 2013). While generally well known heat shock proteins like Hsp70 family chaperones and Hsp90 chaperone subunits were identified, additional associated proteins include nucleotide exchange factors such as FES1 (Bracher & Verghese, 2015), co-chaperones such as STI1-like proteins and Sba1 (Taipale et al., 2010), and protein import facilitators such as Mas5 and Sec72 (Chacinska et al., 2009). These proteins collectively participate in protein folding, stabilization under stress, mitochondrial import, ribosome biogenesis, and telomere maintenance, reflecting the broad cellular response to heat or proteotoxic stress (Balch et al., 2008).

Calcineurin is essential for the regulation of asexual development, stress responses, and pathogenesis in *T. marneffei*, as demonstrated by recent research showing that deletion of the *cnaA* gene results in significant defects in conidiation, germination, morphogenesis, cell wall integrity, and resistance to various stressors (Zheng, Y et al 2020). The “calcineurin” keyword including calcineurin-like phosphoesterase domain-containing proteins, serine/threonine-protein phosphatases, and calcineurin subunit B which are primarily involved in calcium signaling and phosphoregulation (Roy A. et al 2022; Stie, J et al 2008). Overall, showing that our *T. trachyspermus* isolate has specific machinery for production of mycotoxins, especially aflatoxins, as well as key components involved in heat stress response, which could support its classification as a heat-resistant mold.

Conclusions and future work

This study serves as an initial step toward the annotation of the *T. trachyspermus* genome. The proposed workflow offers a foundation that can be further refined and expanded in future iterations. Notably, proteomic features related to mycotoxin production and heat resistance, traits observed in other *Talaromyces* species, were identified. These findings provide a basis for future exploration, including the investigation of specific biosynthetic pathways associated with spoilage in fruit juice matrices.

However, several limitations should be acknowledged. The absence of a repeat masking step may have contributed to a substantial portion of the genome remaining unannotated. Future work should consider integrating repeat masking and evaluating alternative genome assemblers, gene prediction tools, particularly de novo approaches, and functional annotation methods beyond homology-based tools. While UniProt offers a rich repository of curated data, complementary strategies may enhance annotation depth and accuracy. Additionally, the lack of comparative runs limits the ability to benchmark this annotation pipeline against others.

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